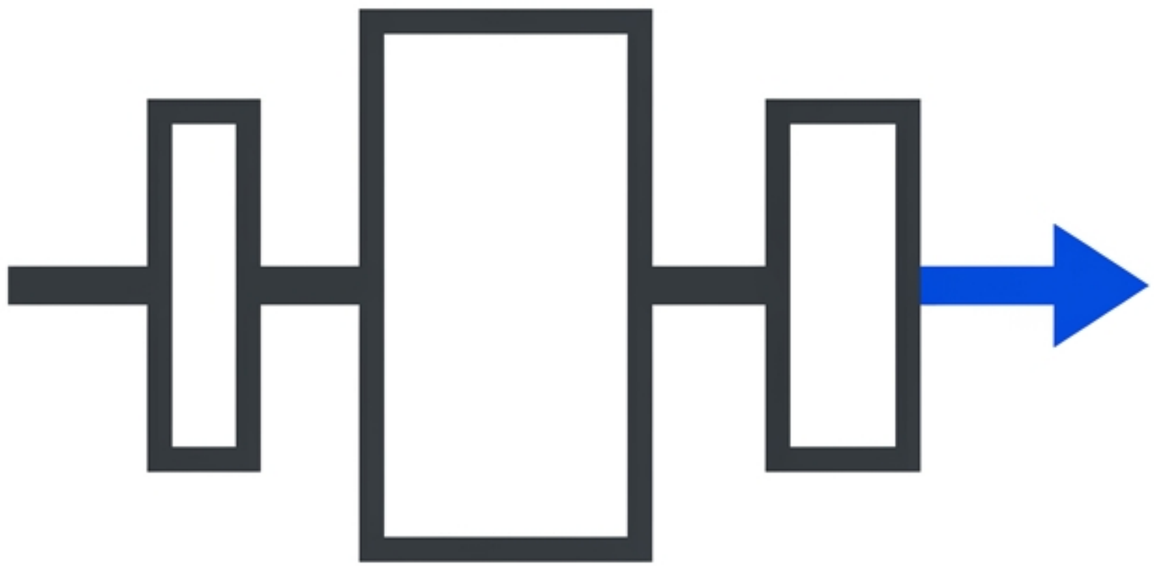
MODULE 16, LESSON 3

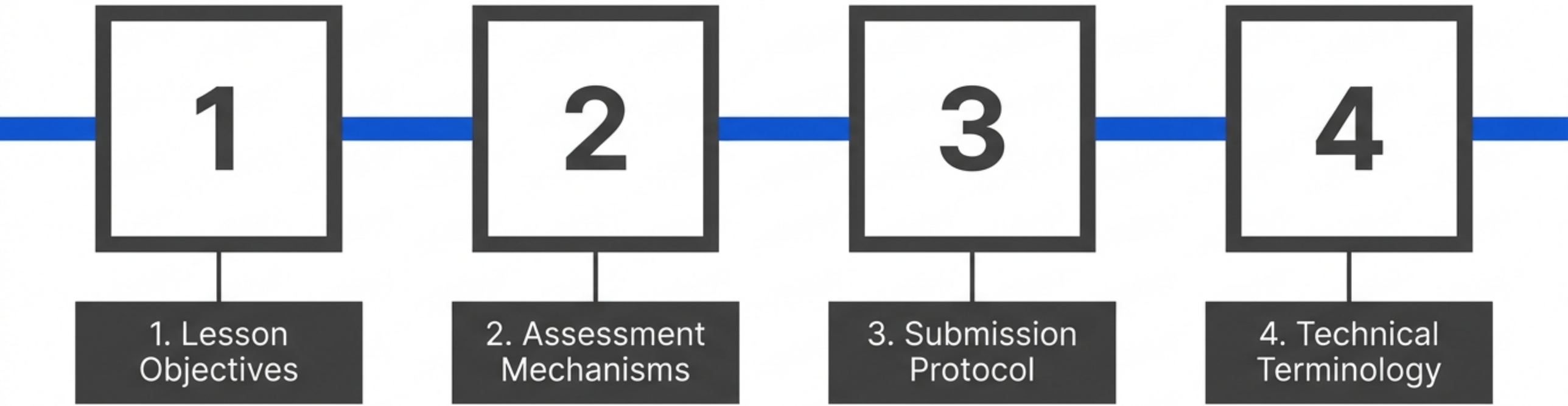
AI 101 for Mechanical Engineers

Submit evidence, not only a browser score



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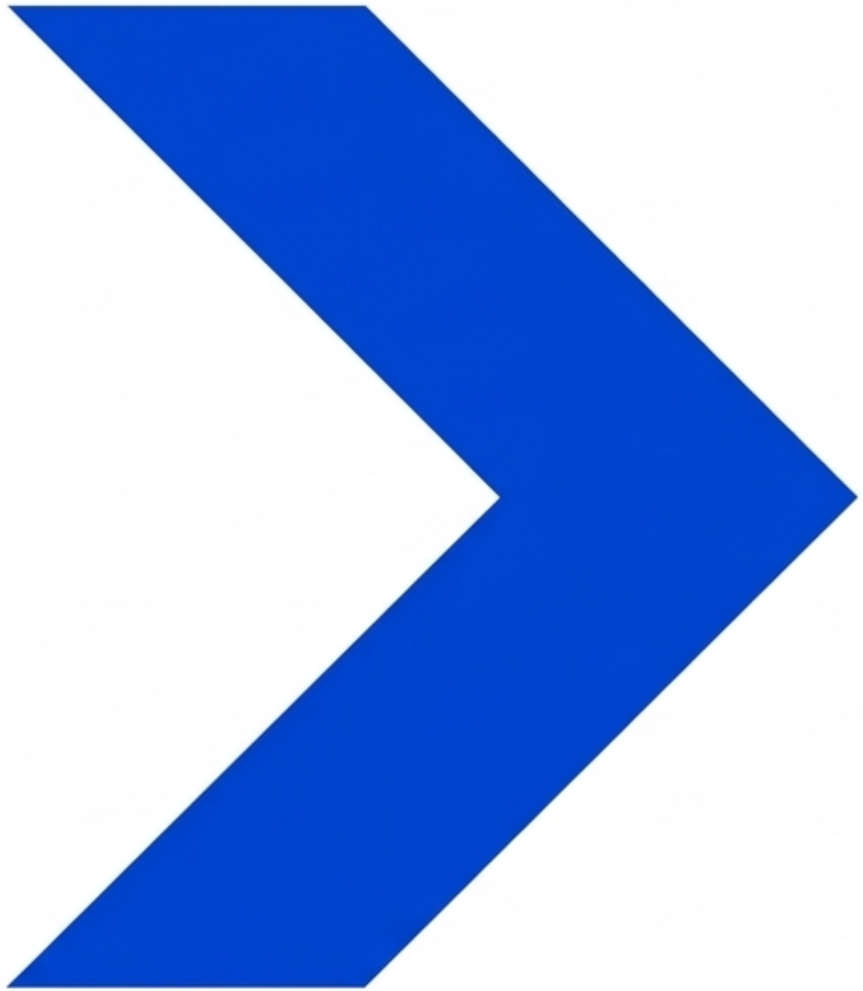
Validation Pipeline Architecture



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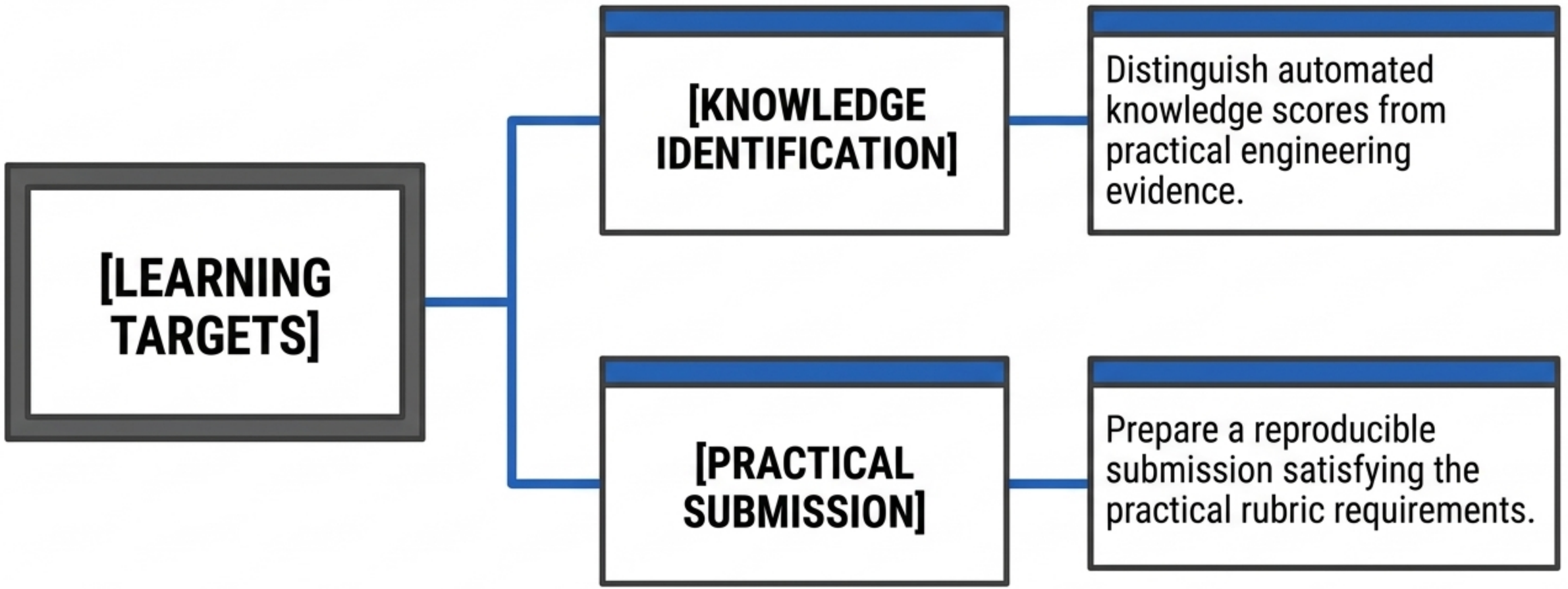
Section 1: Lesson Objectives

Purpose: Establish target competencies for knowledge and practical validation.



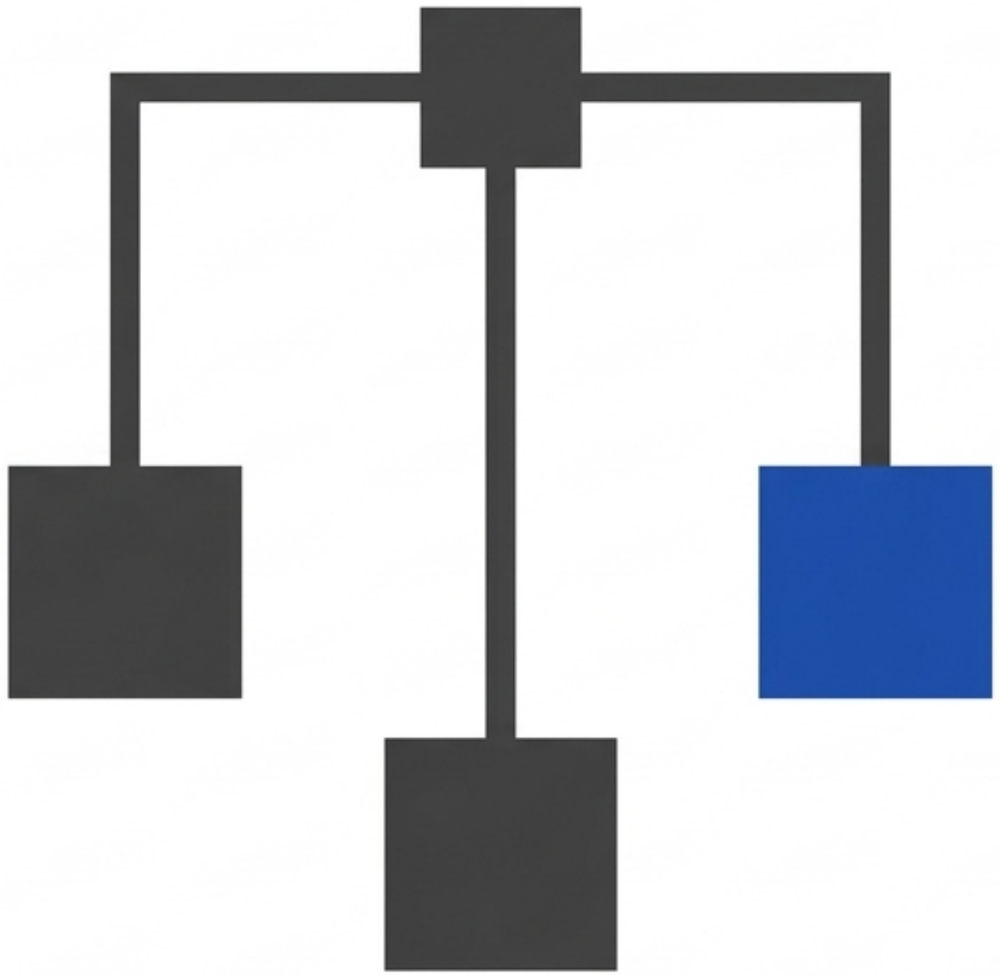
Gemini Notebook

Core Learning Targets



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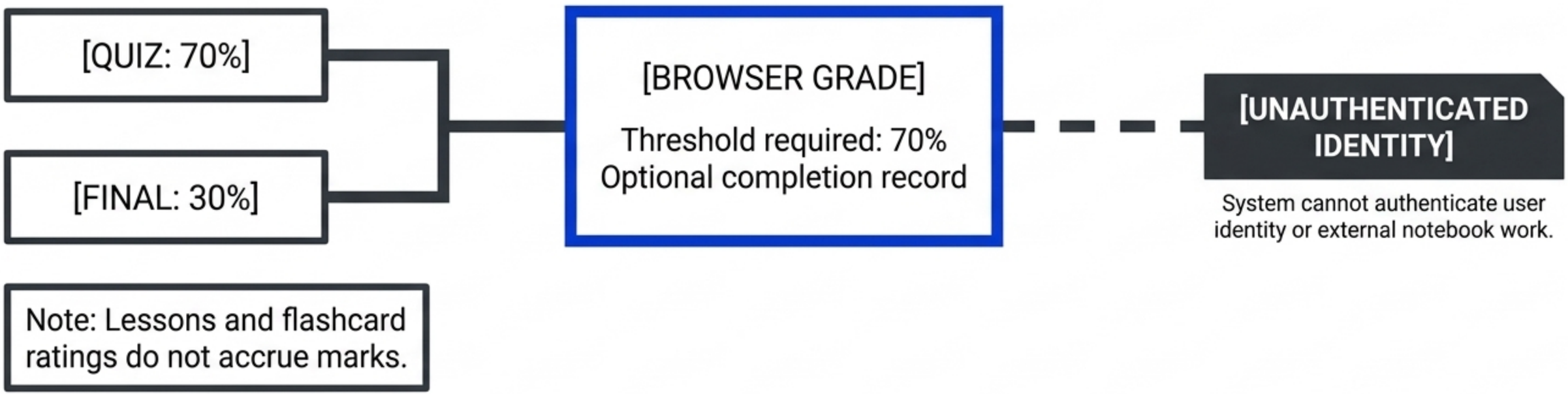
Section 2: Assessment Mechanisms



Purpose: Deconstruct evaluation logic and contrast automated systems against human rubrics.

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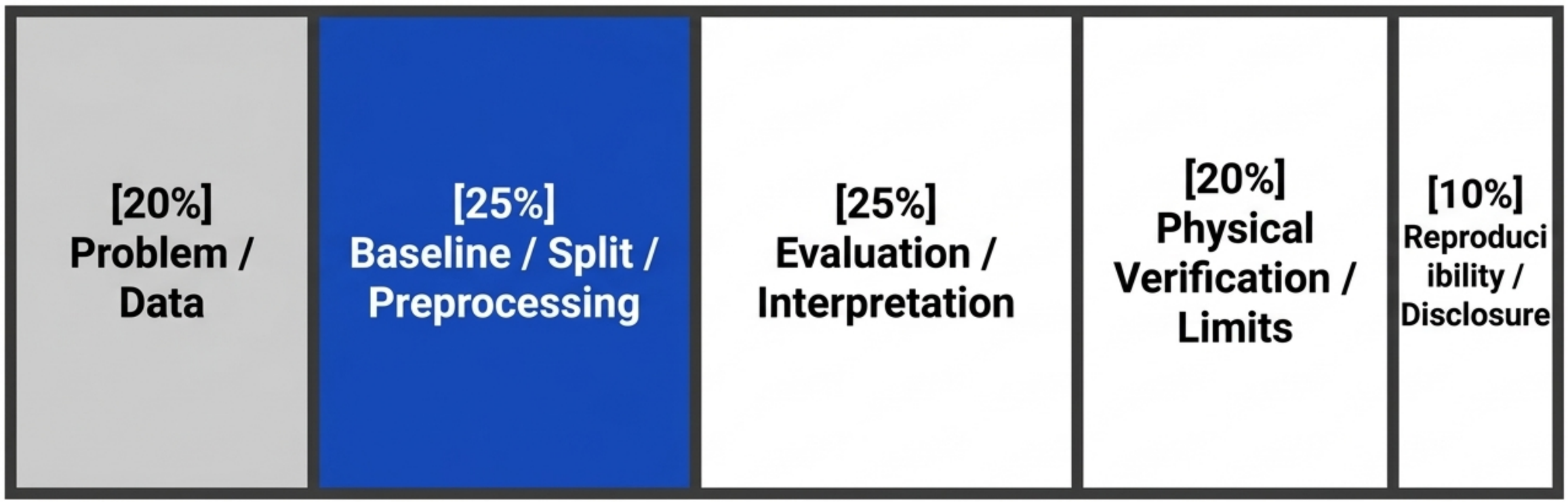
Browser Grade Mechanics



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Rubric Proportional Breakdown

System Constraint: Course systems cannot inspect notebooks automatically. **Practical competence requires separate manual assessment following the 5-part mini-project rubric.**



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Functional Evidence Comparison

Diagnostic Warning: A browser score alone cannot verify practical execution skills.

Evidence Type	Demonstrated Output
Browser Quizzes	Self-reported theoretical understanding
Runnable Notebook	Reproducible practical engineering procedure
Oral Explanation	Rationale and failure-mode comprehension
Practical Rubric	[INSTRUCTOR NODE] or [PEER NODE] validation

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Section 3: Submission Protocol

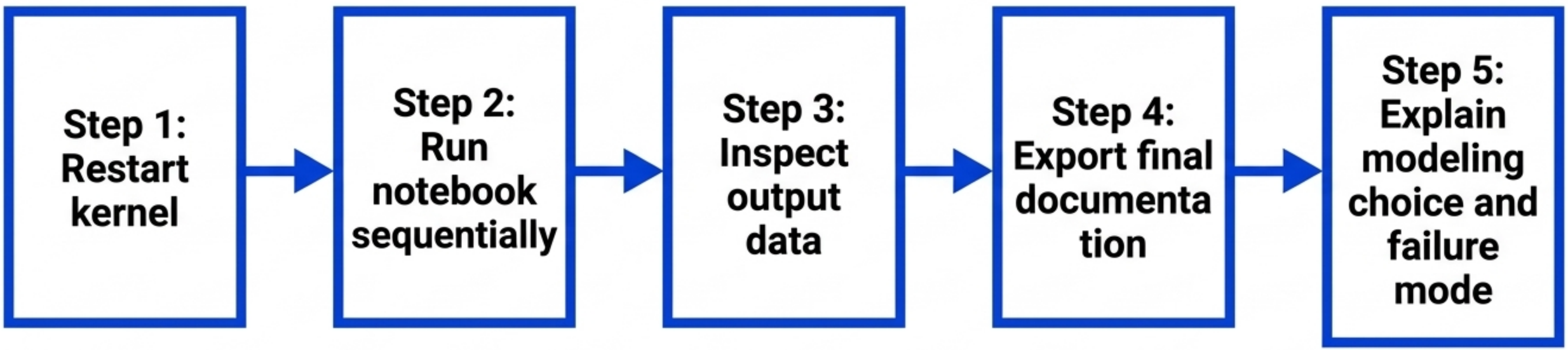
Purpose: Detail the required engineering workflow to produce a defensible output.



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Pre-Deployment Execution Flow

Constraint: Outputs must be fully reproducible from a clean system state. A clean sequential run exposes hidden execution-order dependency errors.



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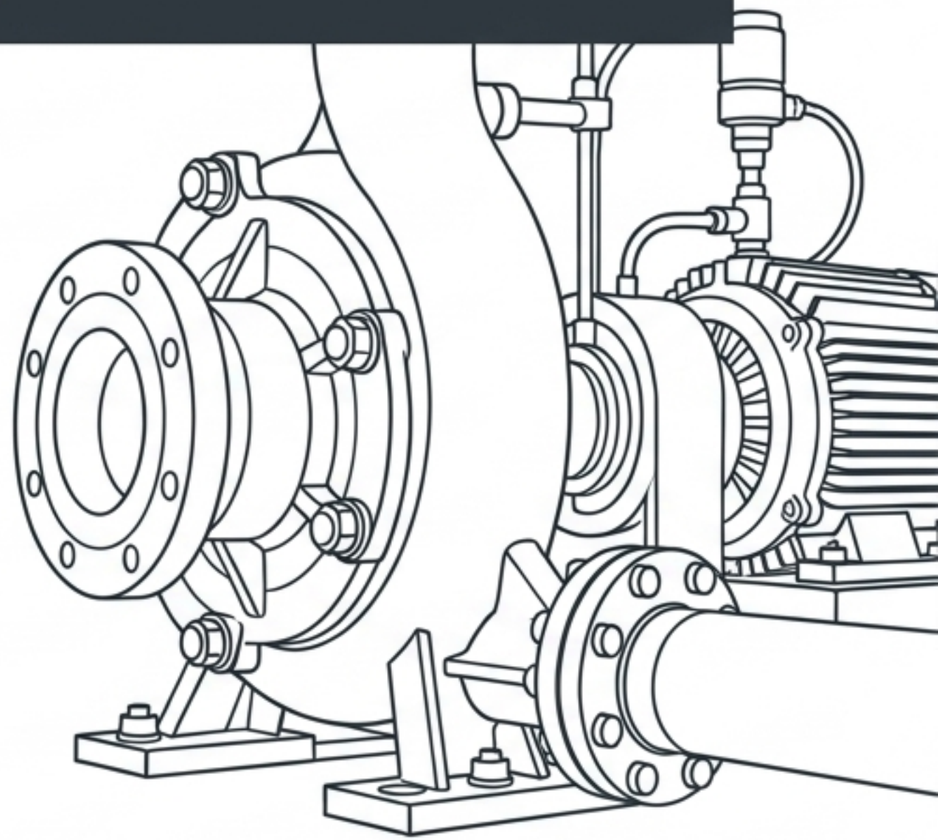
Review and Deployment Limits

Auditing: Reviewers may request direct reproduction of any submitted result.

[BROWSER CERTIFICATE]

Browser certificates do not authorize physical model deployment.

[SYSTEM DEPLOYMENT]

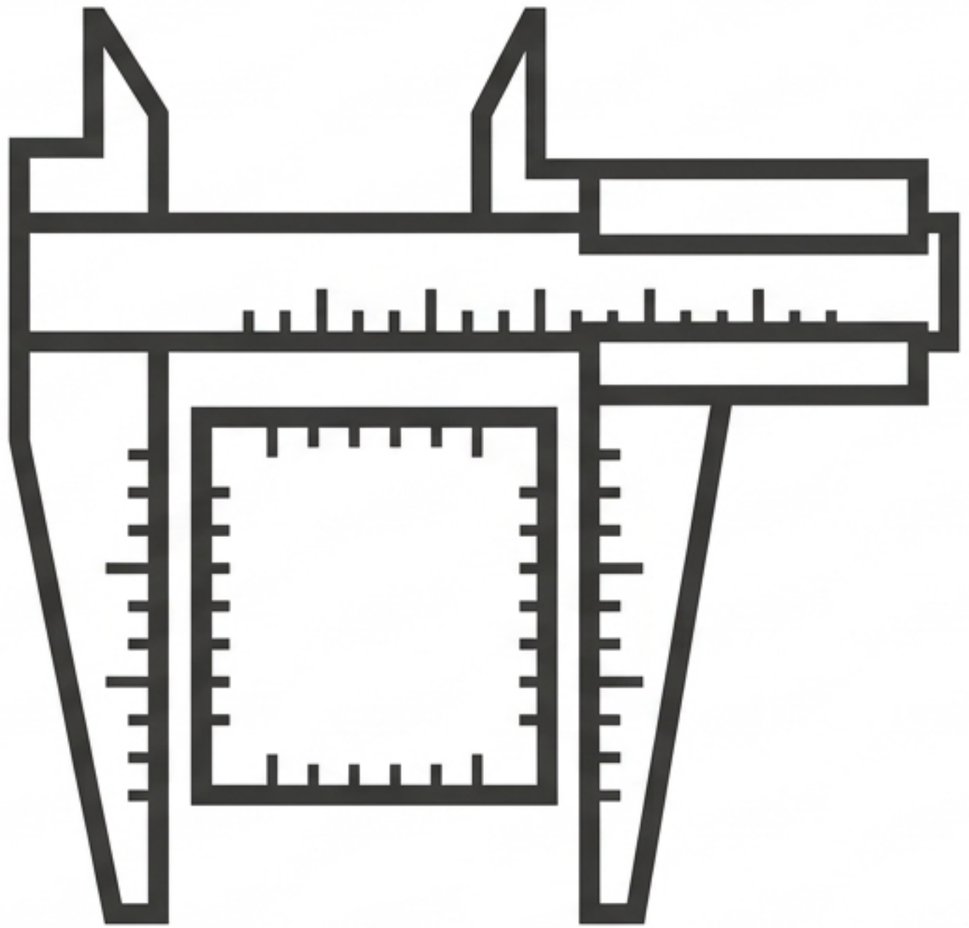


Synthetic evidence does not validate actual mechanical hardware. Real equipment requires external physical validation protocols.

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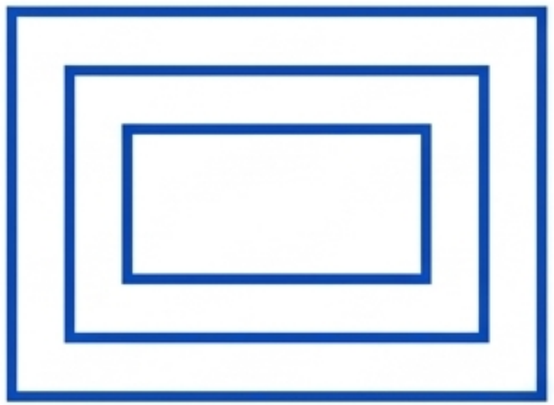
Section 4: Technical Terminology

Purpose: Define operational boundaries and verify diagnostic comprehension.



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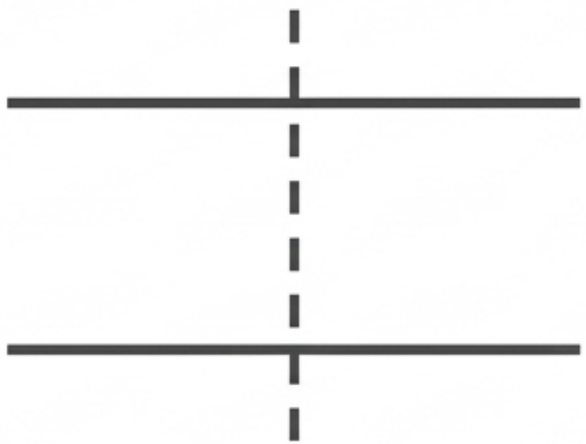
System State Terminology



Reproducibility

Capacity to regenerate procedures using documented inputs and environments.

Execution Example: Executing a computational notebook from a clean system kernel.



Operating Domain

Explicitly evaluated conditions determining system suitability.

Execution Example: Operating within a strictly specified machine and load range.

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Submission Verification Constraints

Axiom: Validation requires physical verification and explicit limitation disclosure.

Flawed Execution	System Reality
Depending on old memory variables	Fails clean kernel reproduction tests
Relying solely on browser grades	Fails to authenticate external work
Using synthetic data exclusively	Fails to validate real mechanical equipment

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Key Points

- **Section 1:** Target competencies distinguish theoretical knowledge from demonstrable practical skill.
- **Section 2:** Automated quizzes assess theory; human rubrics validate practical execution.
- **Section 3:** Valid submissions demand a clean restart, sequential run, and rationale.
- **Section 4:** Engineering reproducibility requires documented inputs and defined operating domains.

Sources: AI 101 Mod 16.3; Google ML Crash Course; scikit-learn Guide.

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